

Project Synopsis

Batch No: F14

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Project Title: CropWise – “Grow smart, farm smart.”

Brief Description:

This project aims to develop an intelligent, AI-based platform to predict crop yields and provide actionable recommendations for small-scale farmers. The system will analyze historical agricultural data, real-time weather information, soil health metrics, and satellite imagery to forecast yields with high accuracy. Using machine learning and deep learning models, the platform will generate insights such as optimal irrigation schedules, fertilizer requirements, and pest management strategies.

The solution will be delivered as a website with a simple, user-friendly dashboard, accessible through any browser without requiring a mobile app or database. Predictions will be processed in real time and displayed instantly, ensuring farmers can make quick, data-driven decisions. The platform will also support multilingual access to ensure wider adoption across different regions.

Objective:

1. Develop an AI platform that predicts crop yields with high accuracy for selected crops and regions.
2. Integrate multi-source datasets including soil properties, weather APIs, and satellite imagery.
3. Build a recommendation system for irrigation, fertilization, and pest control strategies.
4. Design a **website-based dashboard** to visualize predictions and recommendations.
5. Provide multilingual support for easy access by farmers in regional languages.
6. Validate system accuracy using standard evaluation metrics (RMSE, R^2 , MAE).
7. To incorporate multilingual support to overcome language barriers and improve accessibility

Software Requirements:

- Operating System: Windows/Linux/macOS
- Frontend: HTML, CSS, JavaScript (React.js optional for interactivity)
- Backend: Python (Flask or Django)
- Programming Language: Python
- Machine Learning Libraries: Scikit-learn, TensorFlow, Pandas, NumPy, LightGBM/XGBoost, SHAP (for explainability)
- APIs & Services: OpenWeatherMap API (weather data), Sentinel Hub (satellite imagery)
- Version Control: Git & GitHub

Hardware Requirements:

- Processor: Intel i5 / AMD Ryzen 5 or higher
- Memory: Minimum 8 GB RAM (16 GB recommended for model training)
- Storage: 500 MB free space (datasets, models, application files)
- Devices: PC/Laptop for development, any browser-enabled device for usage
- Optional: NVIDIA GPU (e.g., GTX 1660 / RTX 3060) for faster deep learning training

Machine Learning Approach (Model Selection):

The project will primarily use gradient boosting models (LightGBM/XGBoost) as the core prediction method. These models are widely recognized for their high accuracy, fast training, and robustness on tabular agricultural data such as weather, soil, and vegetation indices. They are lightweight, making them ideal for real-time deployment on a website, and also support explainability through SHAP values, helping farmers understand why specific predictions are made.

For extended functionality, the system may also integrate deep learning models:

- CNNs (Convolutional Neural Networks) for extracting features from satellite imagery.
- LSTMs/GRUs (Recurrent Neural Networks) for capturing temporal weather patterns.

Internal Guide (in college)

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External guide (if in Company)

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